Lab 4: Basic Splunk Commands

Learning Objectives

After completing this lab, you will be able to:

- Use the `fields` command to optimize search performance and focus on relevant data
- Create organized tabular reports using the `table` command
- Rename fields for better readability and alignment with business terminology
- Remove duplicate records using the `dedup` command for accurate analysis
- Build complete search pipelines combining multiple commands
- Apply performance optimization best practices
- Create actionable reports for marketing campaign analysis

Prerequisites

- Splunk environment running and accessible
- Sample data loaded (completed Lab 1)
- Field usage skills (completed Lab 3)
- Understanding of search pipelines and command chaining

Lab Scenario

The Marketing team tracks all user sessions related to marketing campaigns. They need a report of all unique user sessions that include purchase actions so they can measure the effectiveness of different campaigns and assign proper value to their marketing investments.

Data Sources

This lab focuses on web application data:

- **Web Application** (sourcetype: `access_combined_wcookie`)
  - Key fields: action, bytes, categoryId, clientip, JSESSIONID, productId, referer, referer_domain, status, useragent, file

> **Data window**: The course dataset covers roughly **26 days** ending at the moment `labs/data/generate_course_data.py` was last run — deliberately sized to sit inside Splunk's **Last 30 days** preset. Set the time picker to **Last 30 days** for every exercise unless a step says otherwise.
>
> Regenerate the data within **4 days** of delivering the class; past that the preset starts clipping the oldest events and the answer-key counts drift.

Task 1: Search for Campaign Purchase Data

Challenge Level

This lab uses a problem-solving approach. Try to complete each step independently before checking the solutions at the end.

Steps:

1. **Navigate to Search Interface**
   - Access Search & Reporting app
   - Set time range to **Last 30 days** (matches the dataset window)
2. **Create Base Search**
   - Search for all web application events with purchase actions and successful status (200)
   - Use: `index=main sourcetype=access_combined_wcookie action=purchase status=200`
3. **Analyze File Field**
   - In the **Interesting Fields**, click on **file**
   - Notice two different files returned: `error.do` and `success.do`
   - Web development team explains:
     - `success.do`: Served when order processed successfully
     - `error.do`: Served when processing error occurs
4. **Filter for Successful Purchases Only**
   - Modify search to include only successful file processing
   - Add: `file=success.do`
5. **Optimize with Fields Command**
   - Use `fields` command to return only: action, JSESSIONID, status
   - Observe if search performance improves
   - Note how field list becomes cleaner

Validation:

- [ ] Base search returns purchase transactions with status 200
- [ ] Identified difference between success.do and error.do files
- [ ] Filtered results to show only successful purchases
- [ ] Applied fields command to focus on relevant data

- [ ] Observed performance improvement

Expected Search Pattern:

```
index=main sourcetype=access_combined_wcookie action=purchase status=200 file=success.do |
fields action, JSESSIONID, status
```

Task 2: Create Marketing-Friendly Reports

Steps:

1. **Convert to Table Format**
   - Replace `fields` command with `table` command
   - Display data in organized tabular format
   - Order: action, JSESSIONID, status
2. **Reorder Columns**
   - Change column order so JSESSIONID appears first
   - New order: JSESSIONID, action, status
3. **Rename for Business Alignment**
   - Marketing team uses "UserSessions" instead of "JSESSIONID"
   - Use `rename` command: `JSESSIONID as UserSessions`
4. **Sort User Sessions**
   - Add `sort UserSessions` command
   - Observe alphabetical ordering
   - Notice duplicate UserSession values
   - Check event count on Statistics tab
5. **Remove Duplicates**
   - Remove `sort` command
   - Add `dedup JSESSIONID` to eliminate duplicate sessions
   - Compare event count before and after deduplication
   - Note: Place dedup early in pipeline for best performance
6. **Final Report Optimization**
   - Remove action and status fields from table display

- Marketing team only needs unique UserSessions list
- Final table should show only UserSessions column

Validation:

- [ ] Successfully created table format display
- [ ] Reordered columns with JSESSIONID first
- [ ] Renamed JSESSIONID to UserSessions
- [ ] Applied sorting and observed duplicate sessions
- [ ] Used dedup to remove duplicates
- [ ] Created clean final report with only UserSessions

Expected Final Search Pattern:

```
index=main sourcetype=access_combined_wcookie action=purchase status=200 file=success.do |
dedup JSESSIONID | table JSESSIONID | rename JSESSIONID as UserSessions
```

Progress Tracking Checklist

Command Mastery

- [ ] Used `fields` command for performance optimization
- [ ] Applied `table` command for report formatting
- [ ] Implemented `rename` command for business alignment
- [ ] Utilized `dedup` command for data accuracy
- [ ] Combined multiple commands in search pipeline

Search Pipeline Construction

- [ ] Built progressive search refinements
- [ ] Optimized command order for performance
- [ ] Created business-ready report format
- [ ] Verified data accuracy at each step

Marketing Analytics

- [ ] Identified successful purchase transactions
- [ ] Filtered out error conditions
- [ ] Created unique user session list
- [ ] Prepared data for campaign value analysis

Performance Optimization

- [ ] Applied fields command early in pipeline

- [ ] Positioned dedup command optimally
- [ ] Minimized unnecessary field processing
- [ ] Created efficient search patterns

Key Concepts Summary

Essential Splunk Commands

`fields` Command

- **Purpose**: Specify which fields to include/exclude from results
- **Performance**: Reduces data processing and improves speed
- **Syntax**: `| fields field1, field2, field3`
- **Best Practice**: Use early in search pipeline

`table` Command

- **Purpose**: Display results in tabular format
- **Formatting**: Creates organized, readable reports
- **Syntax**: `| table field1, field2, field3`
- **Column Order**: Fields appear in order specified

`rename` Command

- **Purpose**: Change field names for clarity or business alignment
- **Syntax**: `| rename old_field as new_field`
- **Multiple Renames**: `| rename field1 as newname1, field2 as newname2`
- **Business Value**: Aligns technical fields with business terminology

`dedup` Command

- **Purpose**: Remove duplicate records based on specified fields
- **Syntax**: `| dedup field1, field2`
- **Performance**: Place early in pipeline when possible
- **Use Cases**: Unique counts, eliminating redundant data

Search Pipeline Best Practices

Command Ordering

1. **Filter Early**: Use search terms and WHERE clauses first
2. **Dedup Early**: Remove duplicates before expensive operations
3. **Fields Selection**: Limit fields early to improve performance
4. **Transform Last**: Apply formatting commands at the end

Performance Optimization

- Minimize data early in pipeline
- Use specific field filters
- Avoid unnecessary sorting operations
- Position dedup strategically

Marketing Analytics Applications

Campaign Measurement

- Track unique user engagement
- Measure conversion rates
- Analyze customer journey stages
- Calculate return on marketing investment

Session Analysis

- Identify unique user sessions
- Track purchase completion rates
- Analyze user behavior patterns
- Measure campaign effectiveness

Real-World Applications

E-commerce Analytics

```
# Track unique customers making purchases by product category
index=main sourcetype=access_combined_wcookie action=purchase status=200
| dedup JSESSIONID, productId
| table JSESSIONID, productId, categoryId
| rename JSESSIONID as CustomerSession
```

Marketing Campaign Performance

```
# Analyze unique sessions per marketing campaign
index=main sourcetype=access_combined_wcookie action=purchase
| dedup JSESSIONID
| stats count by referer
| rename referer as "Campaign Source", count as "Unique Sessions"
```

Customer Journey Analysis

```
# Track customer actions throughout session
index=main sourcetype=access_combined_wcookie JSESSIONID=specific_session
| table _time, action, productId, status
| sort _time
```

Troubleshooting

Common Issues

Fields command not improving performance:

- Verify fields are actually being excluded
- Check if downstream commands require excluded fields
- Consider field extraction overhead vs. benefit
- Monitor search inspector for performance metrics

Table formatting issues:

- Verify field names exist in data
- Check for typos in field references
- Ensure fields have values in result set
- Use `fieldformat` for display formatting if needed

Dedup not removing expected duplicates:

- Verify field names are correct
- Check if fields contain slight variations
- Consider case sensitivity issues
- Use `stats` with `values()` for complex deduplication

Rename command not working:

- Confirm original field name exists
- Check for case sensitivity
- Verify field has values in results
- Use `eval` for complex field transformations

Performance Issues

Slow search execution:

- Move filters earlier in search

- Use more specific search terms
- Limit time ranges appropriately
- Consider summary indexes for repeated searches

Memory or resource errors:

- Reduce field count with `fields` command
- Implement earlier deduplication
- Use more restrictive base search
- Consider search job parallelization

Windows/Mac Compatibility Notes

Windows Users:

- Copy/paste search commands using Ctrl+C/Ctrl+V
- Use Windows-style line breaks if creating saved searches
- PowerShell available for any scripting needs
- Check browser performance with large result sets

Mac Users:

- Copy/paste search commands using Cmd+C/Cmd+V
- Use Unix-style line breaks in search macros
- Terminal available for command line operations
- Monitor browser memory usage with large datasets

Both Platforms:

- Enable browser JavaScript for full functionality
- Clear cache if command completion issues occur
- Use modern browsers for best performance
- Consider browser zoom for table readability

Validation Exercises

Exercise 1: Independent Search Creation

Create a search to find unique user sessions that viewed product category "sports":

Requirements:

- Use web application data
- Filter for "view" actions in "sports" category
- Show unique sessions only

- Display as clean table with renamed fields

Exercise 2: Performance Comparison

Compare performance of two approaches:

1. Dedup early: `base_search | dedup JSESSIONID | other_commands`
2. Dedup late: `base_search | other_commands | dedup JSESSIONID`

Measure and explain the difference.

Exercise 3: Business Report Creation

Create a marketing report showing:

- Unique user sessions per hour
- Only successful purchases
- Formatted for executive presentation

Extension Activities

Advanced Command Combinations

- Use `stats` with `dedup` for complex aggregations
- Combine `eval` with `rename` for calculated fields
- Apply `where` filters after table creation
- Use `sort` with multiple fields and directions

Marketing Intelligence Enhancements

- Add time-based analysis with `timechart`
- Include geographical data with location fields
- Calculate conversion rates using `eventstats`
- Create executive dashboards with formatted tables

Data Quality Improvements

- Implement field validation with `where` commands
- Add data completeness checks
- Create data profiling searches
- Build automated quality monitoring

Cleanup Instructions

1. **Clear Search Interface**

- Remove search criteria from search bar
- Reset time range to default (Last 24 hours)
- Clear any selected fields or filters

2. **Verify Search History**

   - Review Activity > Jobs for completed searches
   - Keep successful searches for reference
   - Clean up any failed or unnecessary searches

3. **Document Key Searches**

   - Save effective search patterns as references
   - Note performance improvements observed
   - Record business insights discovered

Next Steps

This lab covered fundamental Splunk commands for data manipulation and reporting.

End of Content 1

Congratulations! You have completed the Content 1 labs for the Splunk Fundamentals course.

Content 1 Summary

✅ **Lab 1**: Data Loading and Environment Setup ✅ **Lab 2**: Basic Searching with Splunk Search Language ✅ **Lab 3**: Using Fields in Searches ✅ **Lab 4**: Basic Splunk Commands

Coming Up in Content 2

Content 2 will focus on advanced analysis and reporting:

- **Lab 5**: Search Optimization and Troubleshooting (making searches fast)
- **Lab 6**: Transforming Commands (statistical analysis)
- **Lab 7**: Reports and Dashboards (visualization)
- **Lab 8**: Pivot and Datasets (business user tools)
- **Lab 9**: Lookups (data enrichment)
- **Lab 10**: Alerts (proactive monitoring)
- **Lab 11**: Capstone — end-to-end investigation

Solutions Section

Task 1 Solutions

Step 2 - Base Search:

```
index=main sourcetype=access_combined_wcookie action=purchase status=200
```

Step 4 - Filter for Success:

```
index=main sourcetype=access_combined_wcookie action=purchase status=200 file=success.do
```

Step 5 - Add Fields Command:

```
index=main sourcetype=access_combined_wcookie action=purchase status=200 file=success.do |
fields action, JSESSIONID, status
```

Task 2 Solutions

Step 1 - Table Format:

```
index=main sourcetype=access_combined_wcookie action=purchase status=200 file=success.do |
table action, JSESSIONID, status
```

Step 2 - Reorder Columns:

```
index=main sourcetype=access_combined_wcookie action=purchase status=200 file=success.do |
table JSESSIONID, action, status
```

Step 3 - Rename Field:

```
index=main sourcetype=access_combined_wcookie action=purchase status=200 file=success.do |
table JSESSIONID, action, status | rename JSESSIONID as UserSessions
```

Step 4 - Sort Sessions:

```
index=main sourcetype=access_combined_wcookie action=purchase status=200 file=success.do |
table JSESSIONID, action, status | rename JSESSIONID as UserSessions | sort UserSessions
```

Step 5 - Remove Duplicates:

```
index=main sourcetype=access_combined_wcookie action=purchase status=200 file=success.do |
dedup JSESSIONID | table JSESSIONID, action, status | rename JSESSIONID as UserSessions
```

Step 6 - Final Report:

```
index=main sourcetype=access_combined_wcookie action=purchase status=200 file=success.do |
dedup JSESSIONID | table JSESSIONID | rename JSESSIONID as UserSessions
```

Additional Resources

- Splunk Search Command Reference
- Search Performance Optimization Guide
- Marketing Analytics Best Practices
- Data Deduplication Strategies
- Business Intelligence Reporting Techniques